

Study

Convert 2 km/h to m/s. how many ft/s is it?

$$\begin{aligned} 2 \text{ km/h} &= \frac{2 \cancel{\text{km}}}{\cancel{\text{h}}} \left(\frac{1000 \text{ m}}{\cancel{\text{km}}} \right) \left(\frac{\cancel{\text{h}}}{3600 \text{ s}} \right) \\ &= \frac{2000 \text{ m}}{3600 \text{ s}} \\ &= 0.556 \text{ m/s} \end{aligned}$$

* from table 1 ft = 0.3048 m

$$\begin{aligned} 0.556 \text{ m/s} &= \left(\frac{0.556 \cancel{\text{m}}}{\text{s}} \right) \left(\frac{1 \text{ ft}}{0.3048 \cancel{\text{m}}} \right) \\ &= 1.82 \text{ ft/s} \end{aligned}$$

3 sig figs

Convert $300 \text{ lb} \cdot \text{s}$ and 52 slug/ft^3 to SI

from table $1 \text{ lb} = 4.448 \text{ N}$

$$\begin{aligned} 300 \text{ lb} \cdot \text{s} &= 300 \text{ lb} \cdot \text{s} \left(\frac{4.448 \text{ N}}{1 \text{ lb}} \right) \\ &= 1334.4 \text{ N} \cdot \text{s} \\ &= 1.33 \text{ kN} \cdot \text{s} \end{aligned}$$

from table $1 \text{ slug} = 14.59 \text{ kg}$

$$\begin{aligned} 52 \text{ slug/ft}^3 &= \left(\frac{52 \text{ slug}}{\text{ft}^3} \right) \left(\frac{14.59 \text{ kg}}{1 \text{ slug}} \right) \left(\frac{1 \text{ ft}}{0.3048 \text{ m}} \right)^3 \\ &= 26.8 (10^3) \text{ kg/m}^3 \\ &= 26.8 \text{ Mg/m}^3 \end{aligned}$$

$$(50 \text{ mN})(6 \text{ GN}) = [50(10^{-3}) \text{ N}][6(10^9) \text{ N}]$$

Note

$$\text{kN}^2 = (\text{kN})^2 = 10^6 \text{ N}^2$$

$$= 300(10^6) \text{ N}^2$$

$$= 300(10^6) \cancel{\text{N}^2} \left(\frac{1 \cancel{\text{kN}}}{10^3 \cancel{\text{N}}} \right) \left(\frac{1 \cancel{\text{kN}}}{10^3 \cancel{\text{N}}} \right)$$

$$= 300 \text{ kN}^2$$